

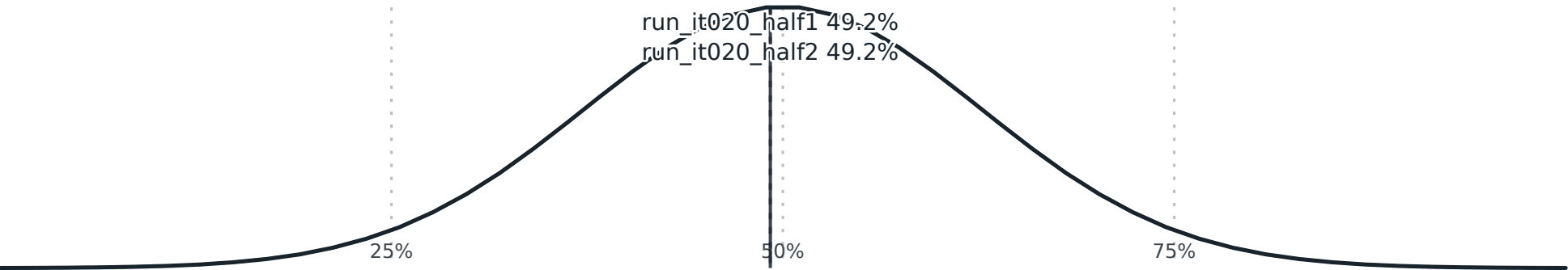
CryoFlow session QC report

Project **β-Galactosidase Tutorial (demo)** · 17 jobs — 17 succeeded · 0 running · 0 failed · 0 waiting.

Pipeline at a glance

- **17** succeeded jobs feed this report's QC sections;
- **0** running · **0** failed — failures stay counted here, never dropped;
- **0** waiting (idle or submitted) — no verdict exists for work that has not run;
- **5m 30s** wall-clock span across **13** runs (first start → last end);
- heaviest step: **Post-processing** at **7m 0s**;
- **52m 0s** of measured compute (windows summed — parallel runs double-count).

Map QC



Map QC summary — run_it020_half1

Job `cmuwipe635000refine3d` · mean-density landscape along **Z** (64 bins).

- **Peak** mean ρ at plane 31 (**49.2%** of depth) — where the specimen's mass concentrates on this axis
- **Trough** at plane 0 (0.0%)
- **Span** across planes: 9.7929 (max 9.7971, min 0.0042)

Comparison maps (1)

Map	Bins	Peak at	Agreement r	Verdict	Shape
run_it020_half2	64	49.2%	1.00	agrees	

Where a comparison line follows the main landscape, the density is consistent between maps; where it parts ways lives noise or masking. Agreement r is the Pearson correlation on the shared 0–100% fraction scale — each terrain self-scaled to its own map's stats (the shape is the signal, not absolute ρ).

Local agreement

Map	Q1 (0-25%)	Q2 (25-50%)	Q3 (50-75%)	Q4 (75-100%)	Weakest	Depth (fraction)	Bands
run_it020_half2	1.00	1.00	1.00	1.00	Q1 (0-25%) (1.00)	0.00-0.25	

A global r can hide a localized betrayal — a map can agree over most of the depth and part ways in a single band (a mask edge, a noise shelf). Each quarter of the shared fraction scale is correlated independently, and the weakest quarter is the address to inspect; the Depth column quotes that address as fractions (the same numbers the viewer's bracket door jumps to). Where the half-maps disagree with each other in one band only, suspect that band, not the reconstruction.

Reading the depth addresses

Every depth in this report is a fraction of the map's thickness along the axis (0 = the front face, 1 = the back face), never an Å position — the landscape is mean density per plane, contour-independent, describing the whole map. The quarter bands are cut by an equal-count knife: each quarter holds the same number of measured planes, so a weak quarter names a real place, not a sparse one. The tables divide the labour — Comparison summarizes each map with one global r; Local addresses (the weakest quarter, quoted as depth); Pairwise corroborates (each pair measured as its own two maps). The Comparison table deliberately carries no Depth column: its address is already quoted in the Local agreement section below — a depth printed twice has two fathers and drifts apart. In the viewer, every one of these addresses is a door: press the chip or the bracket and the plane lands on the band's centre.

Exported from CryoFlow's slice instrument — the mean-density landscape is contour-independent and describes the whole map, not the current isosurface.

Session map inventory

Every job in this session that owns a true 3D volume, newest walk first. The deep profiles above ride the newest owner; this inventory keeps every other owner visible — no map hides below the fold (the t211 lesson: a walk that stops at the first winner leaves the rest of the world unseen). Each row is a door — press it and the page hands you to that job's results (the t210 lesson carried onto the report: a name on a roster should never be a dead end). The Peak column quotes each owner's main map exactly the way the deep report quotes the winner — where the mass concentrates along the shared axis, as a fraction of depth; — means still measuring, never a guess. The Δ winner column reads every peak against the winner's own — the reference row speaks +0.0, its dash obeys the same law, and the lens never guesses; on the page the row farthest from the winner wears the amber edge (a lens, not a verdict — the exported bytes keep the numbers and let the reader judge). The Agreement r column gives every owner the winner's own comparison arithmetic — the same Pearson correlation on the shared 0–100% fraction scale, each owner's main landscape read against the winner's on the finer of the two grids; the reference row lands at 1.00 through the same path, and — means still measuring or a flat field, never a guess. The Weakest column answers the r column's next question — WHERE the shape does not follow: the same

quarter-band cut the deep report's Local agreement table speaks, thinnest band first, Q1-Q4 with the band's own r; the reference row's thinnest quarter is perfect and the cell says so.

Job	Main map	Volumes	Peak	Δ winner	Agreement r	Weakest	Shape
3D auto-refine	run_it020_half1	2	49.2%	+0.0	1.00	Q1 (1.00)	
3D classification	Class 1 map (iter 3)	3	0.0%	-49.2	—	—	—
Topaz Denoise (seeded)	Falcon_2012_06_12-14_33_35_0_denoised	10	0.0%	-49.2	—	—	—
MaskCreate (seeded)	Mask	1	49.2%	+0.0	0.93	Q2 (0.97)	
InitialModel (seeded)	Class 1 map (iter 5)	2	49.2%	+0.0	1.00	Q1 (1.00)	
own motioncorr	Micrograph 14_33_35_0	10	0.0%	-49.2	0.00	Q2 (-0.00)	—
EMPIAR mics import	Micrograph 14_33_35_0	10	0.0%	-49.2	0.00	Q2 (-0.00)	—

The outlier lens is silent here: 3D classification, Topaz Denoise (seeded), own motioncorr and EMPIAR mics import tie for the largest $|\Delta|$ (49.2 vs winner) — a tie for the crown is no crown, so no row wears the amber edge.

Curve verdicts

Every completed job whose workdir carries a curve the charts can draw, newest first. The verdict column is worded over the same interpretation the inspector panels render and the agent's curve tool quotes — one well, three readers; the paper never re-judges what the panels already said.

Job	Curve	Verdict
Topaz Train (seeded)	Picker training	loss 0.9120→0.3660 (falling) · epoch 11: P 72% · R 68%
InitialModel (seeded)	Angular distribution	168 particles · 12/288 bins · $\times 1.8$ — fairly even · hottest r14/t3
Post-processing	FSC	FSC 0.143 at 3.1 Å (0.5 at 3.6 Å) · reported 3.1 Å — final resolution (masked, sharpened)
Post-processing	Guinier	Guinier 36 pts · B-factor -52.4 Å ² (used for sharpening) · covers 2.9–31.6 Å · sharpened curve in plot
3D auto-refine	FSC	FSC 0.143 at ? (0.5 at 5.9 Å)

Job	Curve	Verdict
3D auto-refine	Angular distribution	168 particles · 12/288 bins · ×1.8 — fairly even · hottest r14/t3
3D classification	Angular distribution	168 particles · 12/288 bins · ×2.1 — fairly even · hottest r13/t3
2D classification	FSC	FSC 0.143 at ? (0.5 at 5.9 Å)
2D classification	Angular distribution	240 particles · 15/288 bins · ×2.3 — fairly even · hottest r2/t2
CTF estimation	CTF fit	10 micrographs · mean defocus 1.375 µm · worst fit 8.4 Å (Falcon_2012_06_12-15_30_21_0.mrc)
own motioncorr	Motion drift	10 micrographs · mean drift 5.1 Å · worst 5.8 Å (Falcon_2012_06_12-14_33_35_0.mrc) — early-frames dominate (the stage settles late)

Scheduling sweep

No sweep raced this session — open the HPC queue panel and run *Compare profiles*; the winner's verdict will be bound here verbatim.

Bound from this session's live state — the pipeline glance, the map landscape and the sweep verdict each keep their own provenance. The filename carries the export stamp.